Abhinav Varma Vathadi

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Education

University of Maryland, Baltimore County — Maryland, USA
*Master of Professional Studies in **Data Science** (GPA: 3.86)* — *May 2026*

Experience

Data Scientist — Maryland, USA
UMBC — *Aug 2026 – Present*
- Modeling academic, engagement, and service data with CRISP-DM and ML to target Student Affairs interventions.

Applied AI Engineer Intern — Remote, USA
Anthem Nation — *Jul 2025 – May 2026*
- Cut median inference latency **45%** (4.9s to 2.7s) with **65%** local completion via an AWS Lambda model router.
- Kept workflow fallback to **7.2%** (18/∼250 events) with typed validation, deterministic recovery, and Agents SDK.
- Built incremental Snowflake pipelines across **4 data domains** with Streams/Tasks and versioned Snowpark ML inference.

Data Scientist, Graduate Assistant — Maryland, USA
UMBC (Promoted from Intern) — *Aug 2025 – May 2026*
- Automated recurring EDA, KPI computation, and validation with a Dockerized CrewAI agent for reproducible reporting.
- Modeled engagement across **165,968** student-semester observations with Negative Binomial and zero-inflated regression.
- Validated cohort associations with clustered standard errors and sensitivity checks for Student Affairs planning.

Data Visualization & Reporting Analyst Intern — Maryland, USA
UMBC — *Jun 2025 – Aug 2025*
- Reduced manual survey coding across **2,500+** responses with a human-in-the-loop GenAI/NLP review pipeline.
- Built a Tableau dashboard joining multi-year participation, GPA, and demographic data for cohort analysis.

Data Analyst — Maryland, USA
Retriever Essentials, UMBC — *Sep 2024 – May 2026*
- Lowered stockouts about **15%** by tracking item movement and low-stock patterns with Python, SQL, and Tableau.
- Modeled stockout risk at about **0.75 precision** and shipped a weekly Tableau dashboard for reorder planning.

Associate Software Engineer — Hyderabad, India
MICRON TECHNOLOGY, INC. — *Feb 2023 – Jul 2024*
- Reduced invalid KPI records an estimated **35%** with XGBoost in a Salesforce overhaul and Tableau monitoring.
- Sped document lookup **25%** across **5 production** Einstein AI prompt flows using RAG, Docker, and CI traceability.

Projects

AgentOps EvalBench MCP: LLM Evaluation & Observability | *FastAPI, MCP, Streamlit, PostgreSQL*
- Matched human labels at **87.5%** pass/fail agreement and **0.94** hallucination F1 across 40 RAG evaluation cases.
- Exposed a failure-safe evaluator through FastAPI, Streamlit, Typer, and **6 MCP tools** with persistent run history.

FabPilot AI: Agentic Semiconductor Yield Diagnosis Copilot | *PyTorch, SHAP, LangGraph, Streamlit*
- Auto-accepted **85 of 180** wafer maps at **88.2%** on accepted cases, routing 95 uncertain maps to human review.
- Unified SECOM risk, SHAP, and PyTorch wafer inference via LangGraph in an offline Docker and Streamlit app.

NYC Mobility Big Data Analytics | *Spark, Scala, Hadoop/HDFS, Hive*
- Processed **3.47M** taxi trips and 42,756 traffic rows with PySpark/Spark SQL in a Dockerized HDFS batch pipeline.
- Validated Hive/Scala analytics over **5,658** demand rows, ranking the top 5 pickup zones per hour across all 24 hours.

ClearCast: Weather-Driven Campaign Planner | *LangGraph, FastMCP, GPT-4o-mini, Gradio*
- Generated 5-day campaign plans from live weather data and 4 inputs using OpenWeatherMap and GPT-4o-mini.
- Orchestrated a bounded **10-step** LangGraph/FastMCP workflow with Pydantic validation and checkpointing.

Awards & Publications

- **Outstanding Student Contributions to Assessment Award**, *Student Affairs Assessment Leaders*, International, Jan 2026.
- B. S. Kumar, **V. A. Varma**, S. D. Rani, and A. Nishanth, "Fault detection and classification using machine learning," *Int. J. Res. Appl. Sci. Eng. Technol. (IJRASET)*, vol. 11, no. 4, pp. 3063–3069, Apr. 2023.

Technical Skills

Languages & Data: Python, SQL, Scala, PostgreSQL, Git
Big Data & Distributed: Apache Spark, PySpark, Spark SQL, Hadoop/HDFS, Apache Hive, Parquet
ML & Experimentation: PyTorch, scikit-learn, XGBoost, SHAP, MLflow
AI & Application Frameworks: LangGraph, OpenAI Agents SDK, MCP, FastAPI, Streamlit
Cloud & Analytics: AWS, Snowflake, Docker, GitHub Actions, Ollama